Energy Bills and Manufactured Climate Backlash

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THE LONDON SCHOOL OF ECONOMICS AND POLITICAL SCIENCE

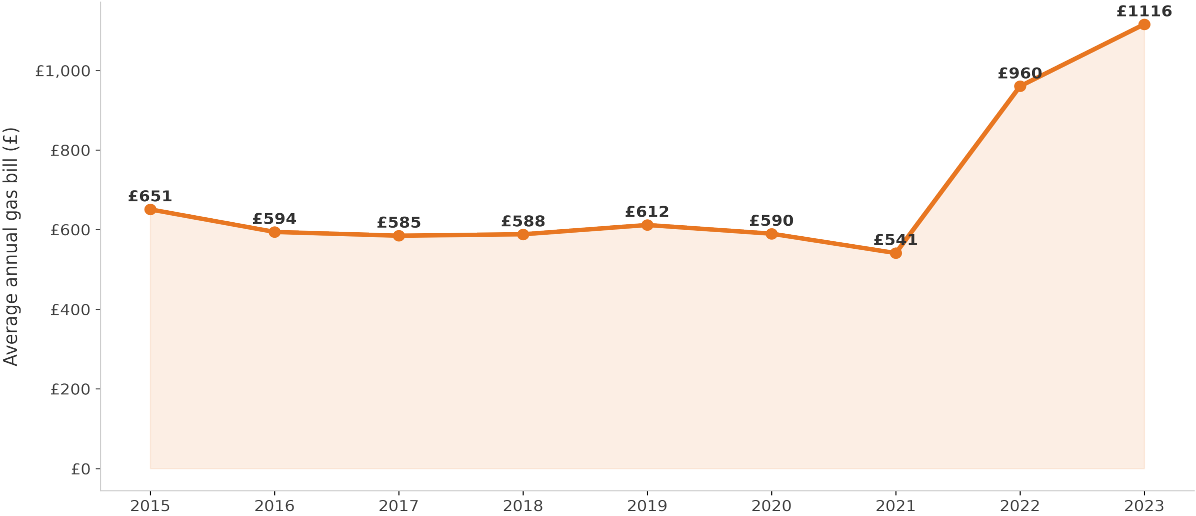
Grantham Research Institute on Climate Change and the Environment

The British Academy


Your latest
energy bill

Average UK Household Gas Bill

UK average annual gas bill including standing charge, weighted by number of meters



Source: OFGEM / BEIS regional gas price data, linked to LSOA meter counts

Green Backlash: The Standard Account

Climate policy imposes direct costs → grievance → vote for the populist radical right

- **Documented across multiple settings:**
  - ↪ Milan low-emission zone → Lega support *(Colantone et al. 2024)*
  - ↪ Dutch gas taxes on renters → PVV support *(Voeten 2024)*
  - ↪ Brown-sector workers → AfD support *(Heddesheimer et al. 2024; Cavallotti et al. 2025)*
- **The common logic:**
  - ↪ A policy generates a cost; the cost generates a grievance; the grievance generates a vote
  - ↪ Backlash bounded by the *material reach* of climate policy

Argument: Manufactured Backlash

*Backlash can also arise when climate policy did **not** cause the costs voters experience*

- **The puzzle:**
  - ↪ The 2021–2023 UK price shock was driven by global gas markets, not Net Zero
  - ↪ Yet Reform UK built its electoral platform on "cut bills by scrapping Net Zero"
- **The mechanism:**
  - ↪ The *shock* supplies the grievance; the *PRR* supplies the causal narrative
  - ↪ Energy bills are multi-causal and ambiguous — voters cannot easily disentangle markets, policy, and firm pricing
  - ↪ Builds on causal narratives in political economy *(Eliaz & Spiegler; Shiller; Kendall & Charles)*
- **The implication:**
  - ↪ The *political* reach of climate policy as a target of grievance exceeds its *material* reach

Research Question

What are the political consequences of rising household energy bills?

Expectations

Rising Household Energy Bills...

- **Supply:**
  ↪ PRR communication increasingly frames rising bills as a consequence of Net Zero

- **Demand:**
  ↪ Households most exposed to higher bills increase support for PRR parties
  ↪ Effects are concentrated among gas-dependent households
  ↪ Exposed households become more likely to oppose environmental protection

Data

Party Communication

Press releases & YouTube

8,669 documents

6 parties, 2019–2026

EPC Ratings

DLUHC Open Data

27m certificates

Pre-2021 efficiency ratings by LSOA

Understanding Society

UKHLS Annual Panel

$\sim$30,000 households

13 waves, 2009–2023

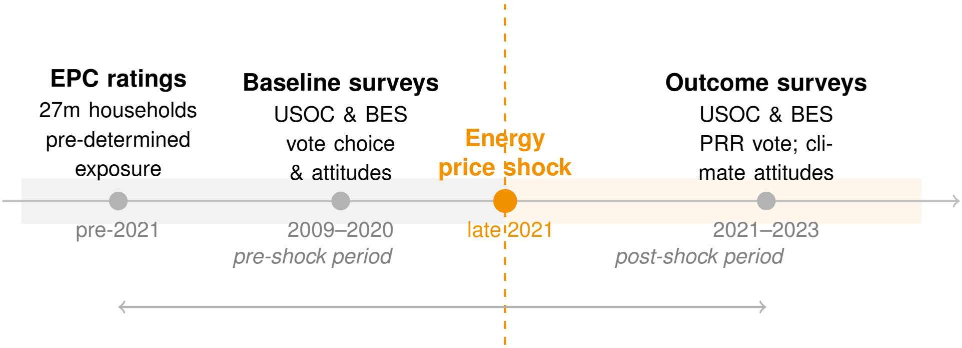
British Election Study

Continuous Online Panel

30 waves

Vote intention & climate attitudes

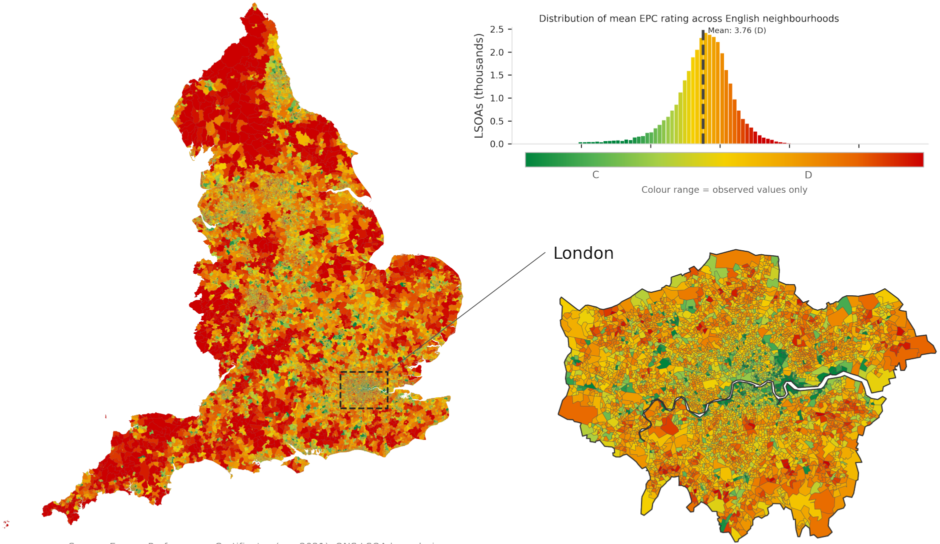
Research Design: a natural experiment



DiD & Event Studies: comparing high vs. low EPC vulnerability areas before and after 2021

Findings

- Energy price shocks fall hardest on communities with older, less efficient housing — creating a **geography of vulnerability**



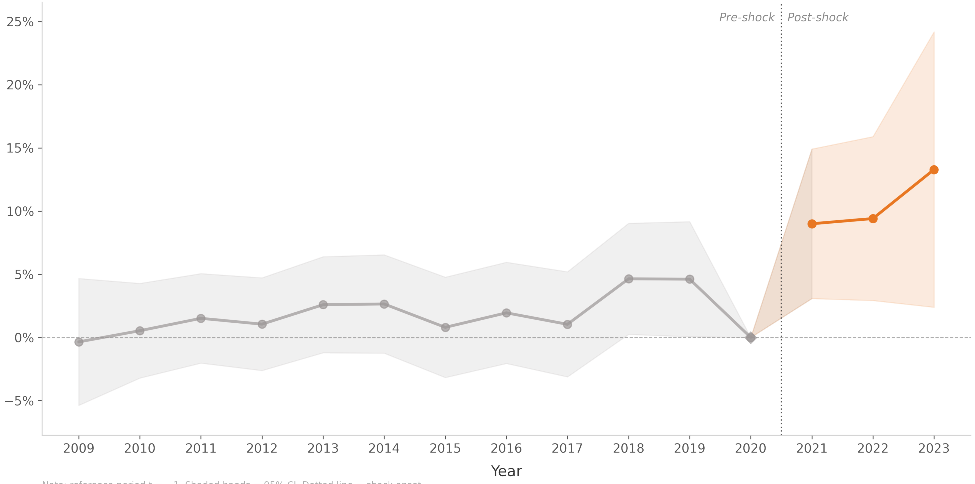
Source: Energy Performance Certificates (pre-2021), ONS LSOA boundaries

Findings

- Energy price shocks fall hardest on communities with older, less efficient housing — creating a **geography of vulnerability**
- Households in energy-inefficient areas paid **disproportionately more** after the 2021 price shock — about **16–22% higher** gas bills per EPC grade of vulnerability

Households in Neighbourhoods with higher EPC ratings experienced larger increases in reported gas bills

Effect of high EPC rating on reported annual gas bill (%) — economic controls



Note: reference period t = −1. Shaded bands = 95% CI. Dotted line = shock onset.
Source: Understanding Society, waves 1-14 (2009–2023).

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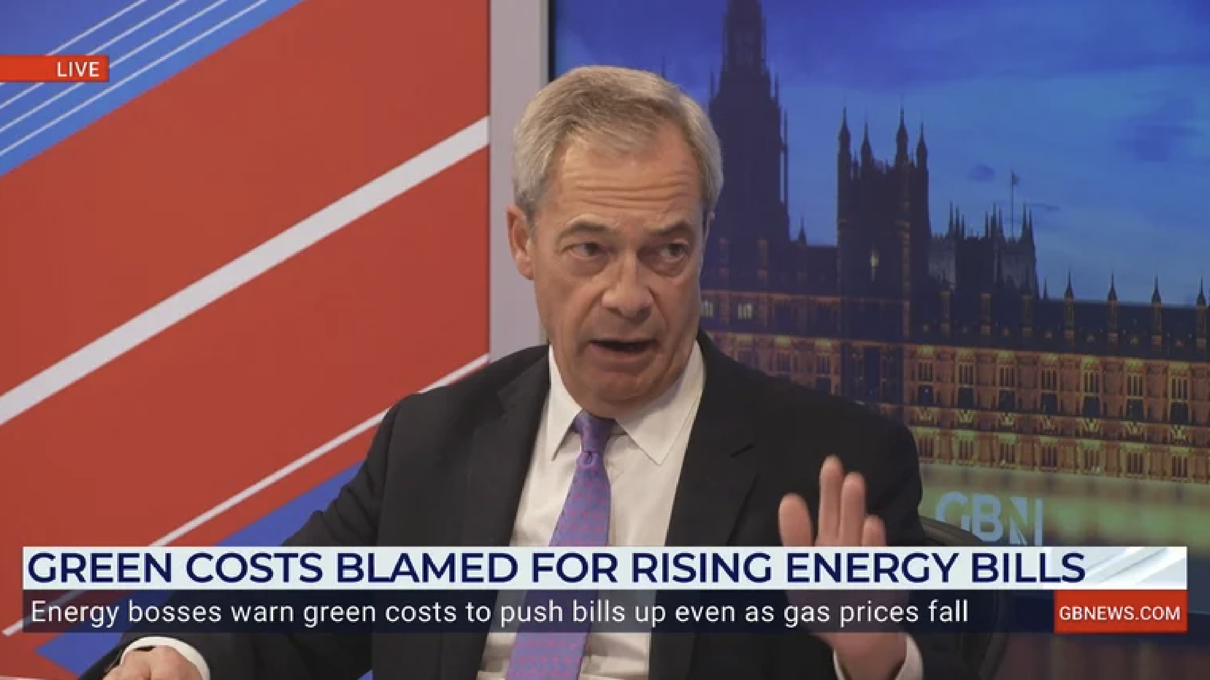
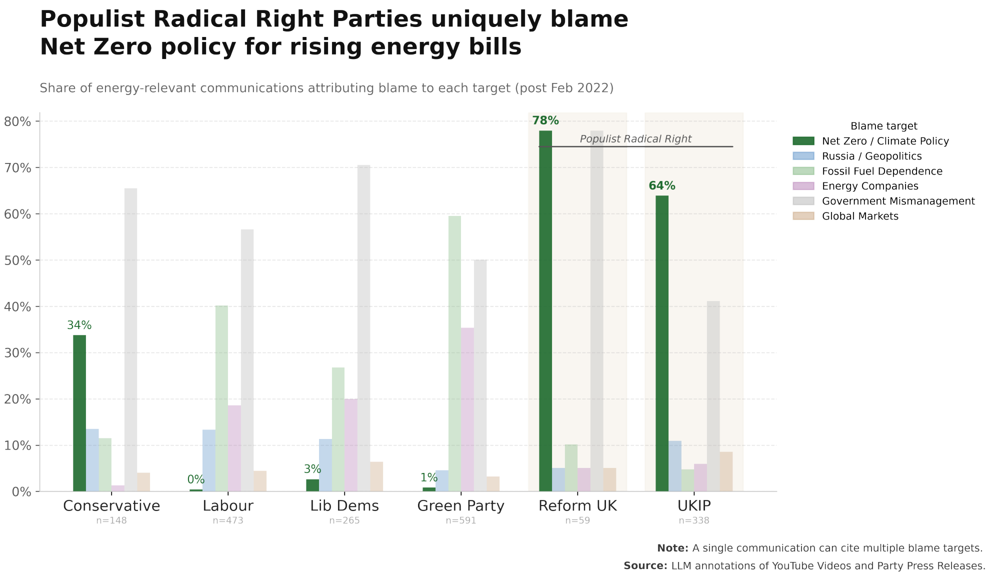
- While mainstream parties blamed Russia and global markets, PRR parties uniquely blamed **Net Zero** — turning energy bills into a weapon against climate policy

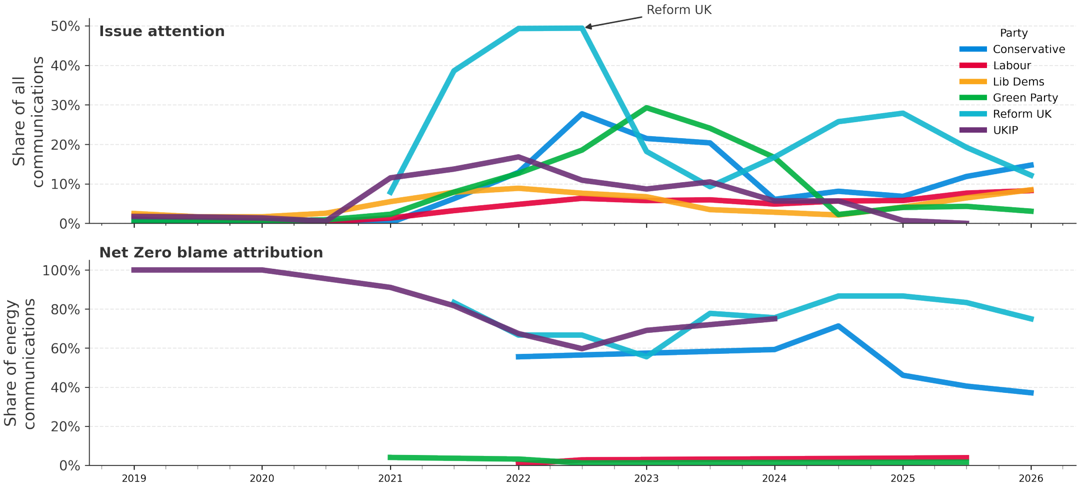

LIVE
GREEN COSTS BLAMED FOR RISING ENERGY BILLS
Energy bosses warn green costs to push bills up even as gas prices fall
GBNEWS.COM



Populist Radical Right Parties uniquely blame
Net Zero policy for rising energy bills
Share of energy-relevant communications attributing blame to each target (post Feb 2022)
80%
70%
60%
50%
40%
30%
20%
10%
0%
34%
0%
3%
1%
78%
64%
Populist Radical Right
Conservative
n=148
Labour
n=473
Lib Dems
n=265
Green Party
n=591
Reform UK
n=59
UKIP
n=338
Blame target
Net Zero / Climate Policy
Russia / Geopolitics
Fossil Fuel Dependence
Energy Companies
Government Mismanagement
Global Markets
Note: A single communication can cite multiple blame targets.
Source: LLM annotations of YouTube Videos and Party Press Releases.

Attention to energy bills increased in 2021, with a large share of PRR communication attributing blame to Net Zero

Monthly smoothed shares: energy-relevant communications (top) and Net Zero blame (bottom)



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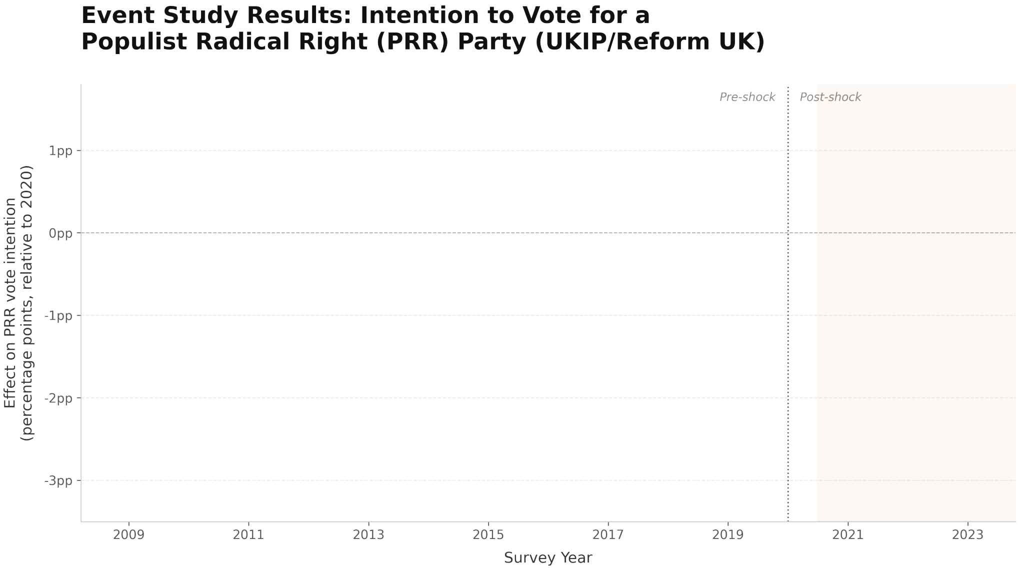
- While mainstream parties blamed Russia and global markets, PRR parties uniquely blamed **Net Zero** — turning energy bills into a weapon against climate policy
- Energy-vulnerable communities became significantly more likely to **support PRR parties** after the 2021 shock — **+0.8 pp per EPC grade** ($\approx$20% relative; $\approx$30% for gas-dependent households) — but **only PRR parties**: Conservatives invoked Net Zero too, with no electoral dividend


Event Study Results: Intention to Vote for a
Populist Radical Right (PRR) Party (UKIP/Reform UK)
Pre-shock
Post-shock
1pp
0pp
-1pp
-2pp
-3pp
Effect on PRR vote intention
(percentage points, relative to 2020)
2009
2011
2013
2015
2017
2019
2021
2023
Survey Year

Energy-inefficient areas increasingly backed Populist Radical Right parties after the 2021 price shock

Effect of high EPC rating on PRR vote intention (percentage points) — three model specifications

Note: reference period 2020. Bars = 95% CI. Dotted line = onset of 2021–22 energy price shock.
Source: Understanding Society (USOC), waves 1–14.

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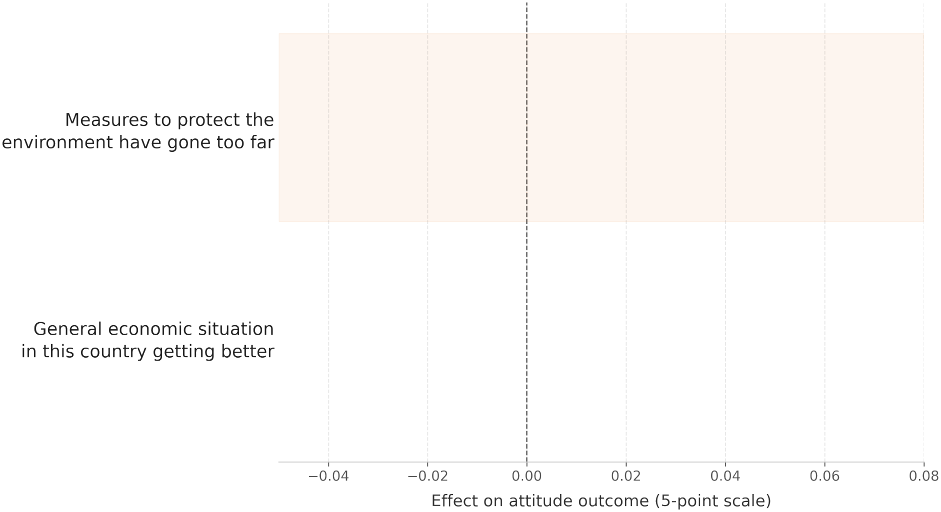
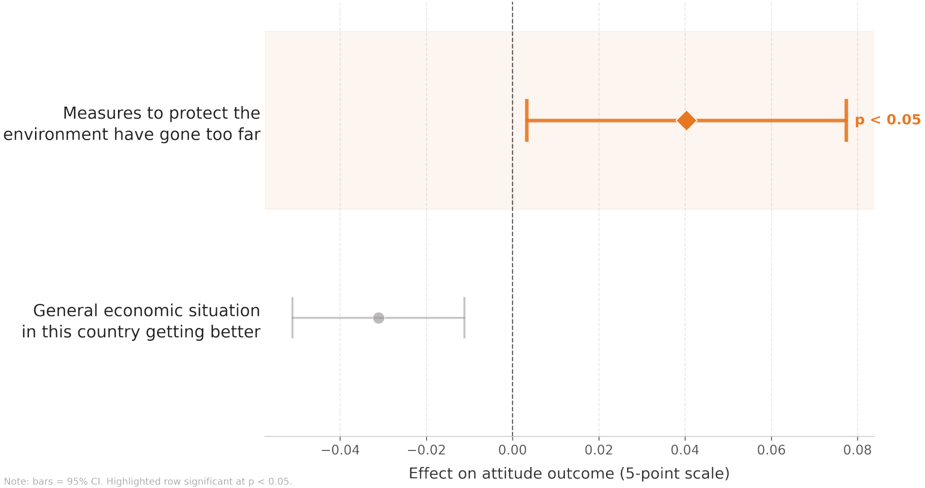
- Energy-vulnerable communities became significantly more likely to **support PRR parties** after the 2021 shock — **+0.8 pp per EPC grade** (≈20% relative; ≈30% for gas-dependent households) — but **only PRR parties**: Conservatives invoked Net Zero too, with no electoral dividend
- The mechanism: rising bills **eroded support for environmental protection** and the green transition

Coefficient Plot Results:
Attitudes about the Environment and Economy



Households most exposed to the energy price shock reduced support for measures to protect the environment

Effect of high EPC rating × post-2021 on BES attitude outcomes — TWFE difference-in-differences



Note: bars = 95% CI. Highlighted row significant at $p < 0.05$.
Source: British Election Study panel (waves 1–30).

Implications: Energy Prices & Climate Policy Support

- Backlash against climate policy can arise **without** climate policy causing the costs — when PRR parties supply the causal narrative

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- The **political reach** of climate policy as a target of grievance *exceeds* its **material reach**

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- Backlash against climate policy can arise **without** climate policy causing the costs — when PRR parties supply the causal narrative
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- Causal narratives are objects of party competition — but require **credibility**: the Conservatives’ null result shows rhetoric without issue ownership does not mobilize voters

Implications: Energy Prices & Climate Policy Support

- Backlash against climate policy can arise **without** climate policy causing the costs — when PRR parties supply the causal narrative
- The **political reach** of climate policy as a target of grievance *exceeds* its **material reach**
- Causal narratives are objects of party competition — but require **credibility**: the Conservatives' null result shows rhetoric without issue ownership does not mobilize voters
- The mechanism travels: any external shock in an energy-adjacent domain (geopolitics, drought, AI demand) is potential ground for manufactured backlash

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Appendix & Robustness

Empirical Specifications

Baseline two-way fixed-effects DiD:

$$\mathrm{Y}_{\mathrm{it}} = \alpha_{\mathrm{i}} + \gamma_{\mathrm{t}} + \beta\,(\text{EPC}_{\mathrm{LSOA}} \times \text{Post}_{\mathrm{t}}) + \mathrm{X}_{\mathrm{it}}^{\top}\boldsymbol{\theta} + \varepsilon_{\mathrm{it}}$$

Event study (dynamic effects & pre-trend test):

$$\mathrm{Y}_{\mathrm{it}} = \alpha_{\mathrm{i}} + \gamma_{\mathrm{t}} + \sum_{\tau \neq \tau_0} \beta_{\tau}\,(\text{EPC}_{\mathrm{LSOA}} \times 1\{\mathrm{t} = \tau\}) + \mathrm{X}_{\mathrm{it}}^{\top}\boldsymbol{\theta} + \varepsilon_{\mathrm{it}}$$

Triple-differences (gas dependence):

$$\mathrm{Y}_{\mathrm{it}} = \alpha_{\mathrm{i}} + \gamma_{\mathrm{t}} + \beta\,(\text{EPC}_{\mathrm{LSOA}} \times \text{GasDep}_{\mathrm{i}} \times \text{Post}_{\mathrm{t}}) + \mathrm{X}_{\mathrm{it}}^{\top}\boldsymbol{\theta} + \varepsilon_{\mathrm{it}}$$

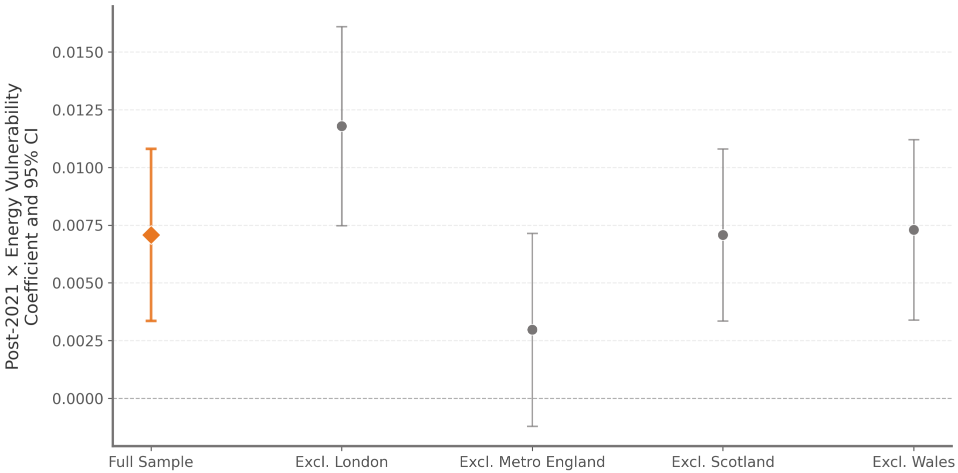
Y_{it}: PRR vote intention (or attitude). α_{i} individual FE, γ_{t} year FE. $\text{EPC}_{\mathrm{LSOA}}$: pre-2021 mean energy (in)efficiency of i's LSOA. $\text{Post}_{\mathrm{t}} = 1\{\mathrm{t} \geq 2021\}$; $\tau_0 = 2020$ (reference). X_{it}: time-varying economic controls. SEs clustered by individual.

back to research design

Robustness: Gas Dependence (Triple-Differences)

Is the effect concentrated where the wholesale gas shock transmitted most directly to bills?

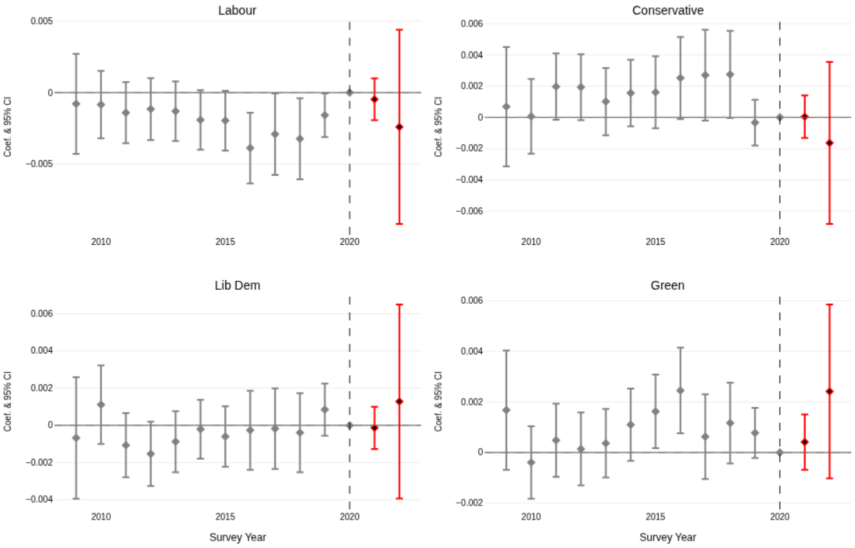
- **Design:** interact energy vulnerability × **pre-shock gas dependence** × post-2021
  ↪ Identifying variation comes from an *independent*, pre-determined source of exposure — guards against vulnerability proxying for other place characteristics
- **Result:** triple interaction **+0.008 to +0.012** (significant across all control sets)
  ↪ ≈1 pp larger PRR support per EPC grade for gas-dependent vs. non-gas households
  ↪ ≈**30% relative** increase over the pre-shock PRR baseline (≈3.6%)



Post-2021 × Energy Vulnerability
Coefficient and 95% CI
0.0150
0.0125
0.0100
0.0075
0.0050
0.0025
0.0000
Full Sample
Excl. London
Excl. Metro England
Excl. Scotland
Excl. Wales

Where the Effect Is Sharpest (Leave-One-Out)

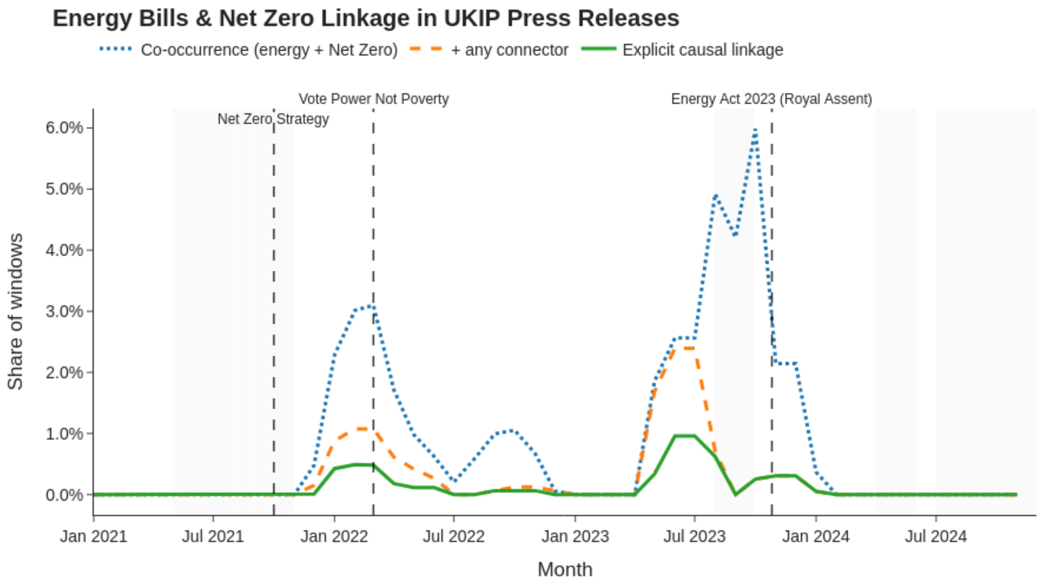
- Dropping Scotland or Wales: **negligible** change (small sample share)
- Dropping **London**: coefficient *increases* (0.0071 $\rightarrow$ 0.0118) — high-efficiency, strongly anti-PRR region suppresses the average
- Dropping **Metropolitan England**: effect attenuates toward zero (0.0030, $\mathrm{p} = 0.16$)
  - $\hookrightarrow$ Post-industrial city-regions combine old, inefficient housing with the communities where Reform UK has made its largest inroads — consistent with the distributive account



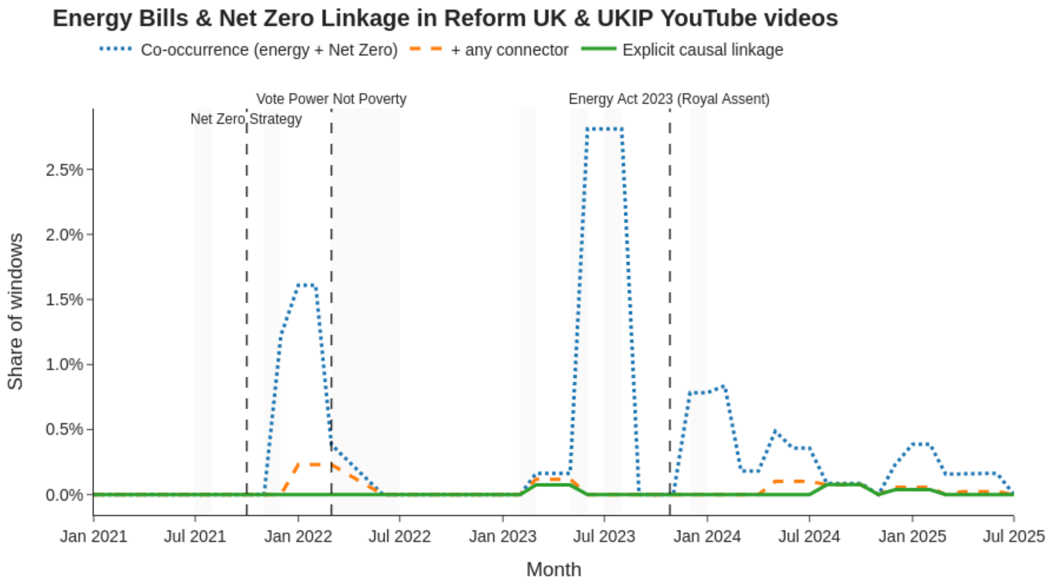
Labour
Conservative
Lib Dem
Green
Coef. & 95% CI
Survey Year
0.005
0
−0.005
0.006
0.004
0.002
−0.002
−0.004
−0.006
2010
2015
2020

The Effect Is PRR-Specific

- Re-estimated for Labour, Conservatives, Liberal Democrats, and the Greens
- Post-2021 coefficients are **small and insignificant** for all four
  - ↪ Energy vulnerability mobilizes support *specifically* for PRR parties — not diffuse electoral discontent
  - ↪ No Conservative dividend despite invoking Net Zero ⇒ what matters is *credible* blame attribution, not rhetoric alone

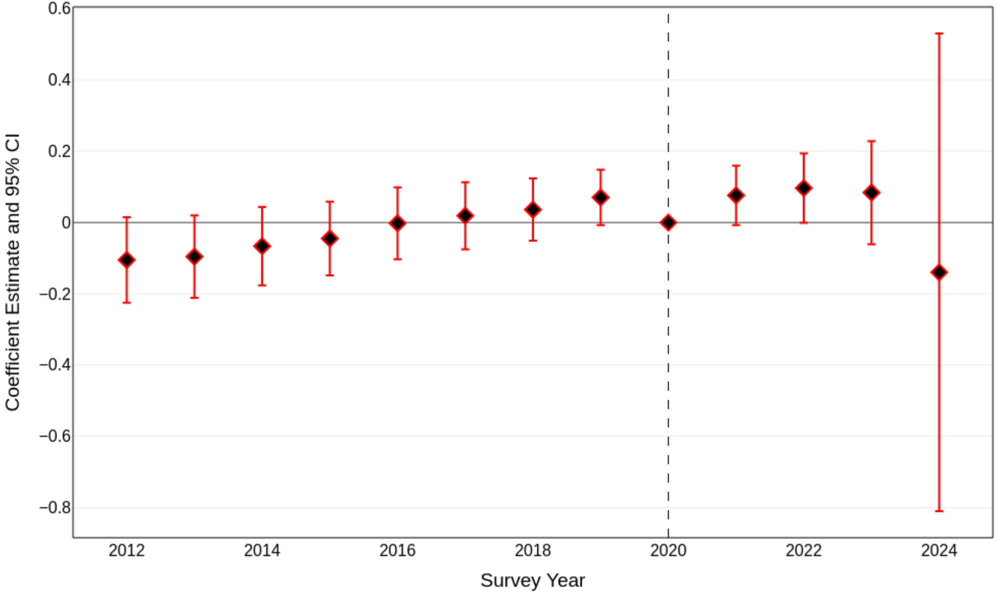


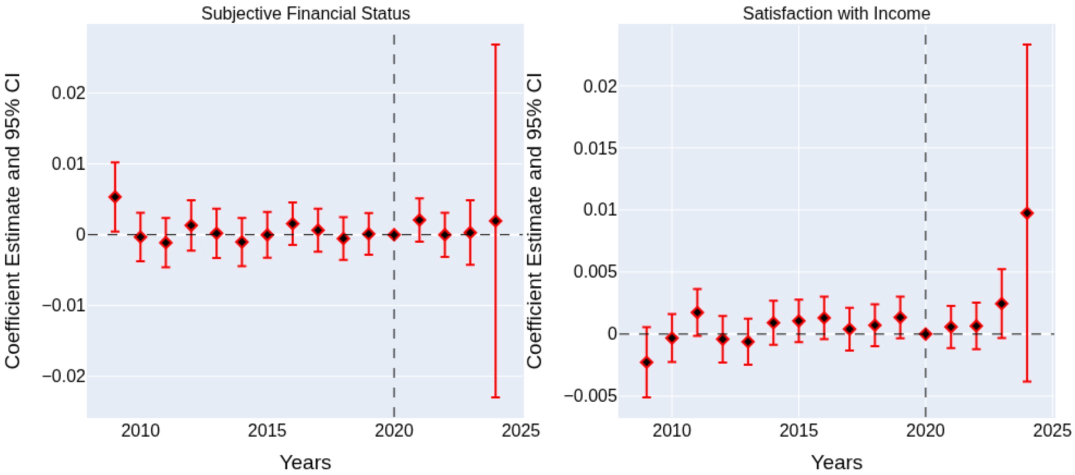
Energy Bills & Net Zero Linkage in UKIP Press Releases
Co-occurrence (energy + Net Zero)
+ any connector
Explicit causal linkage
Vote Power Not Poverty
Net Zero Strategy
Energy Act 2023 (Royal Assent)
Share of windows
6.0%
5.0%
4.0%
3.0%
2.0%
1.0%
0.0%
Jan 2021
Jul 2021
Jan 2022
Jul 2022
Jan 2023
Jul 2023
Jan 2024
Jul 2024
Month


Energy Bills & Net Zero Linkage in Reform UK & UKIP YouTube videos
Co-occurrence (energy + Net Zero)
+ any connector
Explicit causal linkage
Vote Power Not Poverty
Net Zero Strategy
Energy Act 2023 (Royal Assent)
Share of windows
2.5%
2.0%
1.5%
1.0%
0.5%
0.0%
Jan 2021
Jul 2021
Jan 2022
Jul 2022
Jan 2023
Jul 2023
Jan 2024
Jul 2024
Jan 2025
Jul 2025
Month

Text Analysis: Independent Lexicon Check

- Rule-based replication *without* machine-learning annotation
- Counts segments jointly containing: (i) an energy-bill term, (ii) a Net Zero / climate term, and (iii) a **causal connector** (e.g. “because”, “caused by”, “pushing up bills”)
  - ↪ UKIP and Reform UK consistently co-mention bills and Net Zero in causal proximity; the pattern strengthens after the shock
  - ↪ Two methodologically independent approaches **converge** — not an artifact of LLM annotation


0.6
0.4
0.2
0
−0.2
−0.4
−0.6
−0.8
Coefficient Estimate and 95% CI
2012
2014
2016
2018
2020
2022
2024
Survey Year



Source: *Understanding Society Panel (Waves 1-13)*

Placebo Tests: Ruling Out Alternative Channels

- **Income:** energy vulnerability does **not** predict changes in household income
  - ↪ Mechanism runs through the *visibility of the monthly bill*, not labour-market or earnings effects
- **Subjective status:** no association with income satisfaction or subjective financial status, before or after the shock
  - ↪ Not picking up generic grievance / status anxiety
- **Further checks:** LSOA-clustered SEs, binary (above-median) treatment — interaction remains significant at 1%

Limitations

- **Observational panel:** measurement error in EPC ratings attenuates estimates; residential mobility and unobserved local shocks remain possible concerns
- **Partial corpus:** press releases & YouTube only — leaflets, mailing lists, and paid digital ads go unmeasured, so the true intensity of the Net Zero blame frame is likely *higher* than documented
- **No direct Net Zero item:** neither BES nor Understanding Society carries a consistent Net Zero question; the attitudinal mechanism is identified via a broader "environmental protection" item